
AUTOMATED SMART WASTE SEGREGATION SYSTEM WITH LIVE MONITORING AND INTELLIGENT BIN ALERTS

Abstract: Improper waste segregation in urban areas contributes to environmental pollution, overflowing landfills, and poor recycling efficiency. This paper proposes a Smart Waste Segregation System that automates waste detection and categorization using embedded systems and IoT. The system uses sensors to identify metal, wet, and plastic waste and sorts them into appropriate bins using a servo-controlled mechanism. Bin fill levels are tracked via ultrasonic sensors, with data transmitted to a Firebase real-time database through a NodeMCU ESP8266. When a bin is full, users receive notifications via Firebase Cloud Messaging (FCM). An LCD connected to the Arduino Mega displays the waste type and bin status, while a mobile application provides real-time monitoring. This system minimizes manual involvement, improves hygiene, and enables efficient waste handling. Its modular, cost-effective design makes it suitable for residential and community applications, contributing to smart city goals and sustainable urban living.

Keywords: Smart Waste Management, IoT, Embedded Systems, Waste Segregation, Firebase, Real-time monitoring

1 Introduction

Solid Waste Management has emerged as a critical issue with the rapid growth of urban areas and increasing population density. Traditional waste collection and disposal methods involve significant manual effort and are prone to inefficiencies, contamination, and environmental hazards. In particular, improper waste segregation results in the mixing of recyclable and non-recyclable materials, leading to increased landfill volumes and loss of recyclable resources. Furthermore, manual sorting exposes workers to hazardous materials, presenting serious health and safety concerns.

To address these challenges, automation in waste segregation is gaining traction. With advancements in embedded systems, sensor technology, and IoT, it is now feasible to build intelligent waste segregation systems that can automatically identify waste types and ensure their proper disposal. The proposed Smart Waste Segregation System is designed to automate this process using affordable components and scalable architecture. It combines sensor-based detection, mechanical actuation, cloud-based data handling, and user notification mechanisms to create an end-to-end solution for effective waste management.

2 Literature Review

In recent years, technological advancements in embedded systems, sensor networks, and the Internet of Things (IoT) have significantly influenced waste management practices. These innovations aim to tackle the inefficiencies of manual segregation and address environmental and health concerns associated with

improper disposal. A wide range of systems has emerged, each exploring different approaches to monitor, classify, and handle waste more effectively.

One major line of development involves the use of camera-based artificial intelligence (AI) for visual waste classification. These systems often rely on machine learning or convolutional neural networks (CNNs) to identify and categorize waste items based on image recognition. While this method offers high classification accuracy under controlled conditions, it presents practical challenges such as the need for powerful processing hardware, consistent lighting environments, and high-resolution image sensors. Consequently, such solutions may not be viable for low-cost or community-level implementations where affordability and robustness are key concerns.

Other systems focus primarily on bin fill-level monitoring using ultrasonic sensors. These solutions are widely adopted due to their simplicity and low power requirements. They effectively measure the height of accumulated waste and trigger alerts when bins are nearly full. However, these systems lack the capability to differentiate waste types and do not offer any form of automated segregation, limiting their utility to notification-only roles.

A notable example is the work by Gaikwad and Pande (2018), who implemented an IoT-based smart bin that issued alerts when the waste level exceeded a set threshold. While beneficial for monitoring, their system was unable to classify or segregate waste. Sharma et al. (2020) introduced a smart bin that employed a basic moisture sensor to differentiate between wet and dry

waste. However, it lacked the ability to detect metallic objects, resulting in limited classification scope. Similarly, Sinha et al. (2017) developed an Arduino-powered bin that issued alerts via GSM when the container was full, but again, the system did not address the segregation aspect of waste handling.

In contrast, Jain et al. (2022) explored a more recent approach using deep learning models for smart waste detection in urban environments. Their system demonstrated high accuracy in waste type classification using a trained neural network. Nonetheless, it relied heavily on GPU-based processing units and high-bandwidth cloud integration, which may not be feasible for applications in remote or low-resource areas.

These existing systems highlight the trade-offs between cost, accuracy, and functionality. While some excel in detection or alert mechanisms, they often fall short in providing an integrated solution that includes both monitoring and segregation capabilities.

The Smart Waste Segregation System proposed in this paper addresses these gaps by combining multiple detection technologies—metal sensors, moisture sensors, and ultrasonic sensors—with actuator-based sorting and real-time cloud connectivity. The use of both an Arduino Mega and an ESP8266 NodeMCU allows the system to manage local waste classification and global data synchronization simultaneously. It supports both visual and auditory alerts through an LCD display and buzzer, and interacts with a mobile application to provide real-time updates via Firebase and Firebase Cloud Messaging (FCM). This holistic and scalable design bridges the gap between monitoring and segregation, making it suitable for widespread deployment in residential complexes, educational institutions, and public spaces.

3 System Architecture and Methodology

The system is composed of two primary microcontrollers: Arduino Mega and ESP8266 NodeMCU. The waste is placed on a conveyor belt, powered by a 12V DC motor controlled through an L298N motor driver. At the beginning of the conveyor, the metal sensor and moisture sensor detect the nature of the waste item. If the item is metallic, it is directed toward the metal bin. If it is moist, it is classified as wet waste; otherwise, it is considered dry.

An ultrasonic sensor is positioned at the end of the conveyor belt to detect when an object reaches the end. This triggers the belt to stop for 2 seconds, allowing the servo motor to rotate and align the correct bin underneath. The sorting mechanism is implemented using an MG995 servo motor with sufficient torque to rotate a three-bin platform. Each bin is equipped with its own ultrasonic sensor to measure the fill level.

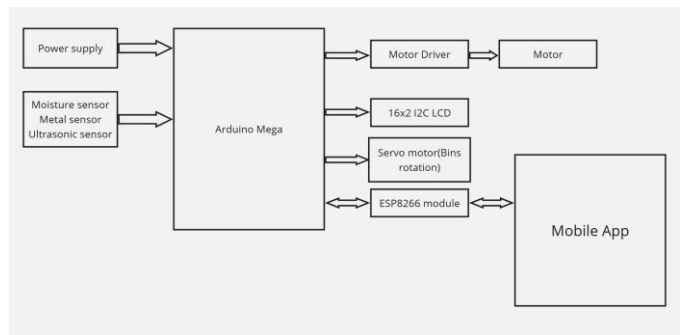


Figure 1 Block diagram of the proposed system

The Arduino Mega handles all sensor readings, motor control, and LCD updates. It sends serial messages to the ESP8266 whenever bin statuses change. The ESP8266 uploads the bin level data to Firebase and, upon detecting a full bin (based on predefined distance thresholds), triggers a notification via Firebase Cloud Messaging. The mobile application, built using Flutter, reflects live bin statuses and displays alerts such as “Dry Bin Full,” “Wet Bin Full,” or “Metal Bin Full.”

To avoid unnecessary power usage and mechanical wear, the conveyor system halts once any bin is full, and resumes only after the bin is emptied. This prevents overfilling and system failure. A buzzer is included to audibly alert nearby users, with distinct tones for waste detection and full-bin alerts.

4 Implementation and Results



Figure 2 Hardware setup of the proposed system

The prototype system as shown in figure 2 was assembled using a modular frame that provided stability and flexibility during development. Three individual bins were firmly mounted to the frame, each representing a distinct waste category—metal, wet, and dry (plastic/paper). All key electronic components, including sensors, microcontrollers, and actuators, were securely attached and organized for ease of testing and maintenance.

A 12V SMPS (Switched Mode Power Supply) was used to power both the Arduino Mega and the DC motor, which drives the conveyor mechanism via an L298N motor driver module. The waste detection setup included a metal sensor, a moisture sensor, and an ultrasonic sensor positioned at the end of the conveyor belt. These sensors were carefully calibrated to distinguish between various waste types and to ensure reliable performance during classification.

The testing process involved placing different commonly found household waste items—such as

aluminum foil, plastic wrappers, and wet tissues—onto the conveyor belt. As the items moved forward, the sensors identified their properties. Based on the detection results, the system automatically sorted each item into the appropriate bin using an MG995 servo motor, which rotated the platform holding the bins to the correct angle.

When an object was detected by the end-of-line ultrasonic sensor, the system temporarily paused the conveyor belt for two seconds, allowing the servo to position the correct bin. After disposal, the conveyor resumed its motion unless a bin was full.

The Arduino Mega managed real-time data collection from all sensors and updated the status display on an I2C LCD module. Messages like “Metal Detected,” “Sorting to Bin 1,” or “Wet Bin Full” were shown during operation. The system was configured to update the fill level of each bin using ultrasonic sensors embedded within them. These readings were transmitted to Firebase Realtime Database approximately every 3–5 seconds through the NodeMCU ESP8266, which handled Wi-Fi communication.



Figure 3 Display the detected waste material

Whenever a bin’s fill level surpassed a preset threshold, the ESP8266 triggered an instant notification using Firebase Cloud Messaging (FCM). A distinct buzzer sound was also activated to alert nearby users. In parallel, the mobile application—developed using Flutter—received real-time updates and displayed them as status bars and push notifications for events like “Dry Bin Full” or “Wet Bin Full.” The interface was designed to be intuitive, allowing users to monitor bin statuses at a glance.

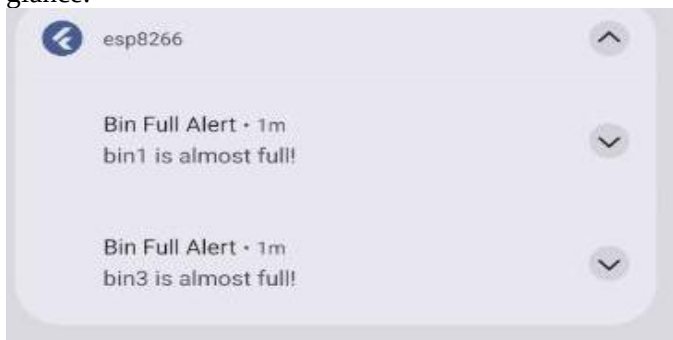


Figure 3 Mobile app notification alert

To prevent mechanical stress and overflow, the system included a safety mechanism: once any bin was full, the conveyor automatically stopped and would resume only after the corresponding bin had been emptied. This

ensured smooth operation, extended hardware lifespan, and minimized maintenance needs.

The prototype performed reliably during testing, confirming the system’s ability to detect, classify, and sort various waste types while maintaining real-time communication and automated alerts.

5 Discussion

The Smart Waste Segregation System developed in this project illustrates the practical potential of embedded systems and IoT technologies in addressing pressing challenges related to urban waste management. By integrating sensor-based waste detection, automated mechanical sorting, cloud-based data transmission, and mobile app interaction, the system provides a comprehensive and real-time solution to reduce manual segregation efforts.

A key strength of the system is its modular architecture, which simplifies maintenance, component replacement, and future scalability. Additionally, the choice of low-cost hardware components, such as the Arduino Mega, NodeMCU ESP8266, and MG995 servo motors, ensures affordability without compromising core functionality. These characteristics make the system well-suited for deployment in a range of environments including residential buildings, universities, smart campuses, and small community clusters.

However, several limitations were observed during the evaluation phase. One of the primary challenges lies in handling mixed-material waste. For instance, items such as food containers with metallic lids and liquid residues may not be classified correctly due to sensor overlap or interference. Similarly, the servo motor, while functionally sufficient, exhibited occasional delays and minor misalignments when sorting under higher mechanical load or prolonged operation. These issues could affect long-term sorting precision.

Another notable challenge is network reliability. Since the system relies on Wi-Fi for data synchronization and notification delivery via Firebase, poor signal strength or intermittent connectivity may lead to delayed or missed alerts. While this does not halt local operation, it limits the system’s effectiveness for remote monitoring.

To overcome these limitations, several enhancements are proposed for future development:

Hybrid classification algorithms that combine sensor fusion and lightweight machine learning models could improve detection accuracy for complex or ambiguous waste types.

Solar-powered modules and energy-efficient firmware could enable operation in outdoor or off-grid environments, improving sustainability.

A web-based dashboard for municipal authorities could extend the system’s utility beyond individual users to support city-wide monitoring and analytics.

Implementing edge computing techniques (e.g., using ESP32 with onboard ML inference) may reduce latency and dependency on cloud processing.

By addressing these limitations and integrating intelligent enhancements, the system has the potential to

evolve into a robust, scalable, and environmentally conscious solution for modern waste management systems aligned with smart city goals.

6 Conclusion

This paper introduced a cost-effective and modular Smart Waste Segregation System that automates the identification, classification, and sorting of household waste using a combination of sensors, embedded hardware, and IoT-based communication. By integrating an Arduino Mega for local control and an ESP8266 NodeMCU for real-time cloud interaction via Firebase, the system successfully performs intelligent waste management with minimal human intervention. Real-time alerts through Firebase Cloud Messaging and an intuitive mobile application further enhance user awareness and encourage hygienic waste disposal practices.

The system's design emphasizes scalability, affordability, and ease of implementation, making it viable for deployment in a variety of urban and semi-urban environments such as residential societies, academic institutions, and public facilities. Its modular structure allows for straightforward upgrades and future expansions.

While the prototype has demonstrated reliable performance in detection, sorting, and real-time monitoring, it also lays the groundwork for a more advanced and autonomous waste management ecosystem. Future iterations may incorporate AI-driven classification, edge computing, renewable energy sources, and integration with smart city infrastructure. Additional features such as a web-based dashboard for centralized data analytics and autonomous robotic waste handling units could significantly broaden the system's impact.

In summary, the proposed system contributes meaningfully to the evolving landscape of smart urban solutions and sustainable living practices, aligning with long-term environmental and technological development goals.

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